Attack Detection and Reconstruction for CPSs Based on Unknown Input Observer and Reachability Analysis

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***Abstract*— This work investigates the problems of attack detection and reconstruction for discrete-time linear cyber-physical systems (CPSs) with unknown-but-bounded disturbances and measurement noise. First, an improved unknown input observer (UIO) with fewer synthesis conditions is proposed to estimate the system state. To enhance the estimation accuracy, a decoupling method is proposed to decouple the disturbance matrix and exclude the effects of disturbances on the error system. Then, to reduce the computational complexity, an improved attack detection method based on the reachability analysis of the residual is proposed, which requires less computation than the existing attack detection methods. Moreover, the impact of disturbances and measurement noise on the process of attack reconstruction and isolation is analyzed. The accuracy of attack reconstruction is improved through the decoupling approach and H_∞ technology. Finally, the simulation results verify the effectiveness and feasibility of the proposed approach.**

***Index Terms*— Attack detection, discrete-time linear cyber-physical systems (CPSs), H_∞ technology, interval estimation, unknown input observer (UIO).**

I. Introduction

RECENTLY, the cyber-physical systems (CPSs) are defined as the physical systems equipped with computing capabilities and communication, which are applied in many areas, such as power systems [1], aerospace [2], and so on. Communication networks make CPSs susceptible to malicious attacks as discussed in [3] and [4]. Consequently, it is essential to improve attack detection and reconstruction strategies to identify malicious attacks on actuators and sensors. Attacks can be classified into the following categories: replay attacks [5], denial-of-service attacks [6], [7], zero-dynamics attacks [8], and false data injection (FDI) attacks [9], [10]. Any attack on the safety-critical CPSs can result in huge financial losses or personal damage.

To prevent such disasters and improve the safety of CPSs, there is a critical need to recognize those malicious attacks. An attack can be considered as a fault within the system. This enables us to utilize the known fault detection approaches, e.g., fault estimation observers [11] and set-theoretic technique [12]. However, the fundamental distinction between an attack and a fault is that the former may be deliberately engineered to evade detection. Chen et al. [13], Pasqualetti et al. [14], and Lu and Yang [15] have presented a substantial amount of research about attack detection. The reachability analysis-based technique has recently attracted significant attention among numerous approaches [16], [4], [17], [18]. These approaches could be divided based on various set representations, such as polytope, ellipsoid, zonotope, and interval. The work [17] used ellipsoidal reachability analysis to investigate the impact of potential stealthy attacks on CPSs. These research findings were further applied to the area of attack detection in [19]. Nevertheless, it was pointed out that in work [20] that ellipsoids were not compact under Minkowski sum and intersection, which required computations leading to excessively high overapproximations. Polytopes can produce more constrained sets, resulting in significant computational costs, such as [21], the interval observer must keep to a cooperative constraint, which restricts its applicability. As pointed in [22], zonotopes showed superior accuracy and more compact representations than ellipsoidal sets, while maintaining lower complexity compared to polytopes. Consequently, further research on attack detection is imperative for attaining superior detection sensitivity and reducing computational burden.

Moreover, reconstructing the attack presents a significant challenge, which led several researchers to explore this

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area [23], [24], [25]. The work [23] reconstructed sparse attacks in CPSs utilizing switched descriptor sliding mode observers. In [24], a novel state observer based on an adaptive switching mechanism was proposed for mitigating sparse sensor attacks in CPSs. The work [25] reconstructed sparse attacks in linear CPSs using a sliding mode observer and a system augmentation technique. The study [26] proposed a new approach based on the projected intermediate estimator for simultaneous reconstruction of sensor and actuator attacks in CPSs, optimizing the algorithms to reduce computational complexity while ensuring the accuracy of attack reconstruction. The research [27] introduced an approach utilizing the zonotopic unknown input observer (ZUIO) for the estimation of system states, as well as the reconstruction and isolation of malicious attacks. ZUIO was applied to detect system faults in [28], [29], and [30]. Therefore, it is significant to develop attack reconstruction methods that achieve both high accuracy and computational efficiency.

Based on the above discussions, this article explores the challenges associated with detecting and reconstructing malicious attacks for discrete-time linear CPSs that are subject to unknown-but-bounded disturbances and measurement noise. The main contributions of our work are outlined as follows.

1) A more general decoupling method is proposed to decouple the disturbance matrix and exclude the effect of disturbances from the error system. Compared with the methods in [31] and [32], the proposed method relaxes the design conditions of the decoupling matrix. In contrast with the methods in [33] and [34], the proposed method has higher state estimation accuracy.
2) To tackle the problem of conservative attack detection thresholds in [4], an improved residual-based attack detection method is proposed to detect attacks. Different from the existing methods in [4], [35], and [36], the proposed attack detection method reduces the computational burden and improves attack detection sensitivity.
3) This work analyzes the impact of system disturbances and measurement noise on attack reconstruction and isolation and provides bounds for reconstruction errors. Compared with the methods in [27] and [37], our approach achieves significantly enhanced accuracy in attack reconstruction.

The rest of this article is organized as follows. Preliminaries and system description are presented in Section II. Section III presents the main results in this article. In Section IV, the simulation results are presented to verify the efficiency of the proposed method. Section V concludes this article.

Notation: $\mathbb{R}^n$ represents a vector with n dimensions and $\mathbb{R}^{m\times n}$ represents a matrix with m rows and n columns. The set of positive integers is $\mathbb{Z}^+$. I_n stands for $n \times n$ identity matrix and $\emptyset$ stands for the empty set. The transpose of a matrix A is represented as A^T and the pseudo-inverse of a matrix A is symbolized by $A^\dagger$. $P \succ 0 (P \prec 0)$ is the means of the positive (negative) definite matrix. Within a symmetric partitioned matrix, the asterisk $*$ is used to represent the transpose of a matrix that is symmetric with respect to its main diagonal. $\|A\|_2$ is denoted as the 2-norm of the matrix A. supp(A) represents the support function of A.

II. Preliminaries and System Description

Definition 1 [38]: An interval vector $\mathbb{I} \in \mathbb{R}^n$ also known as a box is represented as

$$\mathbb{I} = \left[p, q\right] = \left\{x : x \in \mathbb{R}^n, p_i \le x_i \le q_i\right\}$$

where $p_i = [p_1, \ldots, p_n]^T$ and $q_i = [q_1, \ldots, q_n]^T$.

Definition 2 [38]: An i-order zonotope $\mathbb{Z} \subseteq \mathbb{R}^n$ is defined as

$$\mathbb{Z} = \langle p_z, H_z\rangle = \left\langle p_z + H_z z : z \in \mathbf{B}^i \right\rangle$$

where $\mathbb{Z}$ is the affine transformation of the hypercube $\mathbf{B}^i = [-1, 1]^i$, $p_z \in \mathbb{R}^n$ is the center of the zonotope $\mathbb{Z}$, and $H_z \in \mathbb{R}^{n\times m}$ is known as the generating matrix of the zonotope $\mathbb{Z}$.

Definition 3 [39]: For a set $\mathbf{G} \subseteq \mathbb{R}^n$, its interval hull is defined as the smallest interval vector enclosing it, which is represented as

$$\mathbf{G} \subseteq \text{Box}\,(\mathbf{G}) = [c, d] \tag{1}$$

where $c = [c_1, \ldots, c_n]^T$ and $d = [d_1, \ldots, d_n]^T$.

Definition 4 [40]: The Minkowski sum of two sets $\mathbf{Y}$ and $\mathbf{Z}$ is defined as

$$\mathbf{Y} \oplus \mathbf{Z} = \{x + z : x \in \mathbf{Y}, z \in \mathbf{Z}\}$$

where $\oplus$ denotes the Minkowski sum.

Property 1 [40]: For zonotopes, the following equations can be obtained as

$$\mathcal{Z} = \mathcal{Z}_i \oplus \mathcal{Z}_j = \left(p_i + p_j\right) \oplus \left[H_i \quad H_j\right] \mathbf{B}^{i+j} \tag{2}$$

$$L_1\mathcal{Z} = L_1\langle p, H\rangle = L_1 \odot \langle p, H\rangle = \langle L_1 p, L_1 H\rangle. \tag{3}$$

Property 2 [41]: The linear matrix inequality satisfies the following condition:

$$\begin{pmatrix} X & Z \\ Z^T & Y \end{pmatrix} > 0 \tag{4}$$

where X and Y are the symmetric matrices, i.e, $Y = Y^T$ and $X = X^T$. Then, (4) is equivalent to

$$Y > 0, \quad X - ZY^{-1}Z^T > 0. \tag{5}$$

Lemma 1 [42]: For matrices $X \in \mathbb{R}^{a\times b}$, $Y \in \mathbb{R}^{b\times c}$, and $Z \in \mathbb{R}^{a\times c}$, if Y has full column rank, i.e., rank$(Y) = c$, the general solution of the matrix equation $XY = Z$ can be obtained as

$$X = ZY^\dagger + S\left(I - YY^\dagger\right) \tag{6}$$

where $S \in \mathbb{R}^{a\times b}$ is an arbitrary matrix.

Property 3 [16]: For the aim of limiting the order of a zonotope $\mathcal{Z} = \langle \bar{p}_k, \bar{H}_k\rangle$, the reduction operator is represented as

$$\left\langle \bar{p}_k, \bar{H}_k\right\rangle \in \left\langle \bar{p}_k, \hat{H}_k\right\rangle$$

where $\bar{H}_k \in \mathbb{R}^{n\times p}$, $\hat{H}_k = \downarrow_{q,W} (\bar{H}_k)$ and $q \ge n$ denotes the maximum number of columns of matrix $\hat{H}_k = \downarrow_{q,W} (\bar{H}_k)$. First, the columns of $\bar{H}_k$ are ranked by decreasing weighted vector norm, i.e., $\bar{H}_k = [\bar{h}_k^1, \ldots, \bar{h}_k^j, \ldots, \bar{h}_k^p]$ with $\|\bar{h}_k^p\|_W^2 \ge |\bar{h}_k^{p+1}|_W^2$. Then, the set produced by the $p - q + n$ smaller columns is enclosed into a box, i.e., $rs((\bar{H}_k)_<)$, where

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$(\bar{H}_k)_< = [\bar{h}_k^{q-n+1}, \ldots, \bar{h}_k^p]$ and $rs((\bar{H}_k)_<) = \text{diag}(|(\bar{H}_k)_<|\mathbf{1})$. Thus, we have $\downarrow_{q,W}(\bar{H}_k) = [(\bar{H}_k)_>, (\bar{H}_k)_<] \in \mathbb{R}^{n\times q}$ with $(\bar{H}_>) = [\bar{h}_k^1, \ldots, \bar{h}_k^{q-n}]$. If $W = I$, then we represent $\downarrow_{q,W}(\bar{H}_k) = \downarrow_q(\bar{H}_k)$.

Property 4 [33]: For a series of sets $S_i \subset \mathbb{R}^n$, $i = 1, \ldots, m$, the following equation holds:

$$\text{Box}\left(\bigoplus_{i=1}^{m} S_i\right) = \bigoplus_{i=1}^{m} \text{Box}(S_i). \tag{7}$$

In this article, we consider the discrete-time linear system as follows:

$$\begin{cases} x_{k+1} = Ax_k + Bu_k + Ew_k \\ y_k = Cx_k + Fv_k \end{cases} \tag{8}$$

where $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m_1}$, $C \in \mathbb{R}^{p\times n}$, $E \in \mathbb{R}^{n\times m_2}$, and $F \in \mathbb{R}^{p\times q}$ are system matrices. The variables $x_k \in \mathbb{R}^n$, $u_k \in \mathbb{R}^{m_1}$, $y_k \in \mathbb{R}^p$, $v_k \in \mathbb{R}^q$, and $w_k \in \mathbb{R}^{m_2}$ denote the system state, control input, system output, measurement noise, and disturbances, respectively. We assume that x_0, v_k, and w_k are all restricted to their respective zonotopes, which are represented as

$$x_0 \in \langle p_0, H_0\rangle, \quad v_k \in \mathbf{V} = \langle 0, H_v\rangle, \quad w_k \in \mathbf{W} = \langle 0, H_w\rangle \tag{9}$$

where $p_0 \in \mathbb{R}^n$, $H_0 \in \mathbb{R}^{n\times n}$, $H_v \in \mathbb{R}^{q\times q}$, and $H_w \in \mathbb{R}^{m_2\times m_2}$ are known vector and matrices.

When the system (8) in the presence of FDI attacks, the system is shown as

$$\begin{cases} x_{k+1} = Ax_k + Bu_k + Ew_k + D_u a_k^u \\ y_k = Cx_k + Fv_k + D_y a_k^y \end{cases} \tag{10}$$

where the attack sequences $a_k^u \in \mathbb{R}^{m_1}$ and $a_k^y \in \mathbb{R}^q$ are actuator and sensor attacks, respectively. The matrices D_u and D_y are diagonal identity matrices that represent the sensor attack matrix and actuator attack matrices, respectively.

Assumption 1: The matrix pair (A, C) is detectable. The measurement matrix C and the disturbance matrix E are required to satisfy $\text{rank}(CE) = m_2$.

Assumption 2: The following two conditions are necessary for the attack matrices D_u and D_y:

$$\text{span}\{D_u\} \subseteq \text{span}\{B\}, \quad \text{span}\{D_y\} \subseteq \text{span}\{C\} \tag{11}$$

and

$$\text{Null}(D_u) = \text{Null}(D_y) = \emptyset. \tag{12}$$

Remark 1: In practical systems, the attacker injects false data through the physical structure of the system's insecure components. Considering that in the worst scenario, all channels can be designed, i.e., $D_u = B$ and $D_y = C$. Therefore, the attack matrices D_u and D_y ought to satisfy the assumption (11). Then, condition (12) indicates that D_u and D_y have full column rank, allowing for their minimum dimension. This assumption can be found in [16], [18], and [43].

III. Main Results

A. Design of UIO Architecture

In this part, we propose an unknown input observer (UIO) to estimate the state of the system (8). The system state x_k is reformulated as

$$\begin{aligned} x_k &= x_k - NCx_k + NCx_k \\ &= (I - NC)x_k + N(y_k - Fv_k). \end{aligned} \tag{13}$$

Let $d_k = (I - NC)x_k$, (13) can be rewritten as

$$x_k = d_k + N(y_k - Fv_k). \tag{14}$$

Now, the state estimator of the system state x_k can be designed as

$$\hat{x}_k = \hat{d}_k + N(\hat{y}_k - Fv_k) \tag{15}$$

where $\hat{x}_k$ denotes the estimation of the system state x_k, $\hat{y}_k = C\hat{x}_k$, and $\hat{d}_k = (I - NC)\hat{x}_k$. Since $d_{k+1} = (I - NC)x_{k+1}$, taking (8) into d_{k+1}, it gives

$$d_{k+1} = (I - NC)(Ax_k + Bu_k + Ew_k). \tag{16}$$

Then, to exclude the effect of disturbances on the error system, the decoupling matrix $N \in \mathbb{R}^{n\times p}$ is proposed to satisfy the following condition:

$$(I - NC)E = O_{n\times m_2}. \tag{17}$$

Based on (17), we can obtain the equation as

$$NCE = E. \tag{18}$$

Therefore, according to the Lemma 1, the solution of the decoupling matrix N is represented as

$$N = E(CE)^\dagger + S_0\left(I_p - (CE)(CE)^\dagger\right) \tag{19}$$

where S_0 is an arbitrary matrix.

Substituting (17) into (16) and adding the feedback term $L_a(y_k - \hat{y}_k)$, according to $\hat{y}_k = C\hat{x}_k$, we can obtain

$$\hat{d}_{k+1} = (I - NC)(A\hat{x}_k + Bu_k) + L_a(y_k - \hat{y}_k) \tag{20}$$

where L_a is the gain of the UIO. Combining (15) and (20), the proposed UIO can be expressed as

$$\begin{aligned} \hat{x}_k = {} & (I - NC)A\hat{x}_{k-1} - L_a c\hat{x}_{k-1} + L_a y_{k-1} \\ & + (I - NC)Bu_{k-1} + Ny_k. \end{aligned} \tag{21}$$

According to $e_k = x_k - \hat{x}_k$, the error system can be obtained as

$$\begin{aligned} e_{k+1} = {} & [(I - NC)A - L_a c]e_k - L_a Fv_k - NFv_{k+1} \\ & + (I - NC)D_u a_k^u - L_a D_y a_k^y - ND_y a_{k+1}^y. \end{aligned} \tag{22}$$

To obtain a more simplified form of the UIO, we define

$$I_n - NC = T \tag{23}$$

where $T \in \mathbb{R}^{n\times n}$.

Subsequently, the error system is rewritten as

$$e_{k+1} = (TA - L_a c)e_k + B_e f_k + C_e a_k \tag{24}$$

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where

$$B_e = [-L_aF \quad -NF]$$
$$C_e = \begin{bmatrix} TD_u & -L_aD_y & -ND_y \end{bmatrix}$$
$$f_k = \begin{bmatrix} v_k^T & v_{k+1}^T \end{bmatrix}^T$$
$$a_k = \begin{bmatrix} a_k^{uT} & a_k^{yT} & a_{k+1}^{y\ T} \end{bmatrix}^T.$$

We utilize the H_∞ technology to design UIO, such that the error system e_k is robust against the measurement noise v_k. Now, we proposed the following Theorem 1.

Theorem 1: Under attack-free condition, given a scalar $\gamma > 0$, if there is a positive definite matrix $P \in \mathbb{R}^{n\times n}(P \succ 0)$, matrices $W \in \mathbb{R}^{n\times p}$ and $Y \in \mathbb{R}^{n\times(n+p)}$ satisfying the following condition:

$$\begin{bmatrix} I_n - P & * & * & * \\ 0 & -\gamma^2 I_p & * & * \\ 0 & 0 & -\gamma^2 I_p & * \\ \xi_1 & \xi_2 & \xi_3 & -P \end{bmatrix} \prec 0 \tag{25}$$

where

$$\xi_1 = PA - PE\Theta^\dagger CA - Y\Psi CA - WC$$
$$\xi_2 = -PE\Theta^\dagger F - Y\Psi F$$
$$\xi_3 = -WF$$

and T, N, L_a can be obtained by

$$T = I_n - E\Theta^\dagger C - S_0\Psi C \tag{26}$$
$$N = E\Theta^\dagger + S_0\Psi \tag{27}$$
$$L_a = P^{-1}W \tag{28}$$

where

$$S_0 = P^\dagger Y, \quad \Theta = CE, \quad \Psi = I_{n+p} - \Theta\Theta^\dagger$$

then (21) is a robust observer of the system (8) and $\|e_k\| < \gamma\|f_k\|$ is satisfied.

Proof: The Lyapunov function is defined as

$$V(k) = e_k^T P e_k, \quad P > 0. \tag{29}$$

Then, the criterion function is chosen as

$$J = \sum_{k=0}^{\infty} \left(\Delta V(k) + e_k^T e_k - \gamma^2 v_{k+1}^T v_{k+1} - \gamma^2 v_k^T v_k \right) \tag{30}$$

where $\Delta V(k) = V(k+1) - V(k)$. In order to achieve the expected performance, we require $J < 0$. Then, one can obtain

$$\Delta V(k) + e_k^T e_k - \gamma^2 v_{k+1}^T v_{k+1} - \gamma^2 v_k^T v_k < 0. \tag{31}$$

The above condition (31) is equivalent to the following condition:

$$\begin{bmatrix} \Omega_{11} & * & * \\ \Omega_{21} & \Omega_{22} & * \\ \Omega_{31} & \Omega_{32} & \Omega_{33} \end{bmatrix} \prec 0 \tag{32}$$

where

$$\Omega_{11} = (TA - L_ac)^T P (TA - L_ac) + I_n - P$$
$$\Omega_{21} = -(NF)^T P (TA - L_ac)$$
$$\Omega_{31} = -(L_aF)^T P (TA - L_ac)$$
$$\Omega_{22} = (NF)^T PNF - \gamma^2 I_p$$
$$\Omega_{32} = (L_aF)^T PNF$$
$$\Omega_{33} = (L_aF)^T PL_aF - \gamma^2 I_p.$$

According to the Schur complement lemma, we have

$$\begin{bmatrix} I_n - P & * & * & * \\ 0 & -\gamma^2 I_p & * & * \\ 0 & 0 & -\gamma^2 I_p & * \\ \xi_1 & \xi_2 & \xi_3 & -P \end{bmatrix} \prec 0 \tag{33}$$

where $\xi_1 = P(TA - L_ac)$, $\xi_2 = -PNF$ and $\xi_3 = -PL_aF$. Based on the Lemma 1, the matrices T and N can be obtained by (19) and (23), respectively. Ultimately, following the bounded real lemma [44], (25) guarantees the stability of the error system (22) and the condition of $\|e_k\| < \gamma^2\|f_k\|$ is satisfied. □

Remark 2: The H_∞ performance of the UIO is optimized by solving the optimization problem as

$$\min \ \gamma^2$$
$$\text{s.t. } (25). \tag{34}$$

Theorem 1 presents an approach to define the variable $S_0 = P^{-1}Y$, which eliminates the need for manual selection of matrix S_0. This strategy improves the accuracy in determining matrices T, N, and L_a, while concurrently enhancing the robust of the error system e_k.

B. Attack Detection Based on Zonotopes

In this subsection, an improved approach for detecting malicious attacks based on the residual will be presented. When the system (8) is under attack, the corresponding UIO is transformed into

$$\hat{x}_k^a = TA\hat{x}_{k-1}^a - L_ac\hat{x}_{k-1}^a + L_ay_{k-1}^a + TBu_{k-1} + Ny_k^a \tag{35}$$

where x_k^a represents the system state under FDI attacks and $\hat{x}_k^a$ stands for the estimation of the system state in the presence of attacks.

Therefore, the error system under FDI attacks is represented as follows:

$$\begin{aligned} e_k^a &= \overline{A}e_{k-1}^a + TD_ua_{k-1}^u - ND_ya_k^y - L_aD_ya_{k-1}^y \\ &\quad - L_aFv_{k-1} - NFv_k \\ &= \overline{A}^k e_0^a - \overline{A}^k ND_ya_k^y - \sum_{i=0}^{k-1} \overline{A}^i L_aFv_{k-i-1} \\ &\quad - \sum_{i=0}^{k-1} \overline{A}^i NFv_{k-i} + \sum_{i=0}^{k-1} \overline{A}^i TD_ua_i^u - L_aD_ya_0^y \\ &\in \langle p_k^a, H_k^a \rangle \end{aligned} \tag{36}$$

with

$$\begin{aligned} p_k^a &= \sum_{i=0}^{k-1} \overline{A}^i TD_ua_i^u - L_aD_ya_0^y + \sum_{i=1}^{k-1} \overline{A}^i (N + L_a) D_ya_i^y \\ &\quad + \overline{A}^k ND_ya_n^y \\ H_k^a &= \begin{bmatrix} \overline{A} \downarrow_q (H_{k-1}^a) & -L_aFH_v & -NFH_v \end{bmatrix} \\ \overline{A} &= (I - NC)A - L_ac \end{aligned}$$

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where $\downarrow_q (H^a_{k-1})$ is the reduction operator, ensuring that the number of columns never exceeds q at any time instant k.

Subsequently, the residual under FDI attacks can be derived as

$$\varepsilon^a_k = Ce^a_k + Fv_k + D_y a^y_k \in \langle \widetilde{p}^a_k, \widetilde{H}^a_k \rangle \tag{37}$$

where

$$\widetilde{p}^a_k = CAp^a_{k-1} + D_y a^y_k$$
$$\widetilde{H}^a_k = \left[C \downarrow_q (H^a_k) \quad FH_v \right].$$

Therefore, the center of the residual $\widetilde{p}^a_k$ only can be affected by FDI attacks. When the system is attack-free, the center of residual is zero. Now, the following Theorem 2 is given to detect FDI attacks.

Theorem 2: Under the attack-free condition, the residual ε_k can be represented as $\varepsilon_k \in \langle 0, \widetilde{H}^a_k \rangle$, where

$$\widetilde{H}^a_k = \left[C \downarrow_q (H^a_k) \quad FH_v \right] \tag{38}$$

with

$$H^a_k = \left[C\bar{A} \downarrow_q (H^a_{k-1}) \quad -L_aFH_v \quad -NFH_v \right]. \tag{39}$$

Proof: Since $e_0 = x_0 - \hat{x}_0$ and $x_0 \in \langle \hat{x}_0, H^a_k \rangle$, we can obtain that $e_0 \in \langle 0, H^a_0 \rangle$. Then, based on (22), we have

$$e_{k+1} \in \bar{A} \langle 0, H^a_k \rangle \oplus (-NF) \langle 0, H_v \rangle \oplus (-L_aF) \langle 0, H_v \rangle.$$

Thus, we can get $e_{k+1} \in \langle 0, H^a_{k+1} \rangle$. Since the residual $\varepsilon_k = Ce_k + Fv_k + D_y a^y_k$, such that $\varepsilon_k \in C\langle 0, H^a_k \rangle \oplus F\langle 0, H_v \rangle$. Hence, we can gain $\varepsilon_k \in \Pi^a_k = \langle 0, \widetilde{H}^a_k \rangle$.

Subsequently, when the system is attack-free, the system residual satisfies $\varepsilon_k \in \Pi^a_k$. □

Based on Theorem 2, it can be concluded that the zonotope Π^a_k always includes ε_k when the system is attack-free. When the condition $\varepsilon_k \notin \Pi^a_k$ is satisfied, the system is attacked. Therefore, the subsequent attack detection approach based on the zonotope Π^a_k is proposed as

$$\begin{cases} \varepsilon_k \in \Pi^a_k, & \text{attack-free} \\ \varepsilon_k \notin \Pi^a_k, & \text{attacked.} \end{cases}$$

The aforementioned attack detection approach needs to determine the geographic link between a point and a zonotope. However, the configuration of the zonotope Π^a_k may be complex, potentially resulting in a serious computing cost. To reduce computational complexity, we present an improved attack detection method grounded in the reachability analysis of the system residual, which is articulated in the subsequent Theorem 3.

Theorem 3: For the system (8), the interval estimation of the system residual $[\underline{\varepsilon}_k, \overline{\varepsilon}_k]$ can be gained with $\hat{x}_0 = p_0$. When $k = 0$, the interval estimation of the system residual ε_0 can be obtained as

$$\left[\overline{\varepsilon}_0, \underline{\varepsilon}_0\right] = \text{Box}\left(\langle 0, CH_0 \rangle\right) \oplus \text{Box}(F\mathbf{V}). \tag{40}$$

When $k = 1$, the interval estimation of the system residual ε_1 can be represented as

$$\left[\overline{\varepsilon}_1, \underline{\varepsilon}_1\right] = \text{Box}\left(C\overline{A}\langle 0, H_0 \rangle\right) \oplus \text{Box}\left(-CL_aF\mathbf{V}\right) \oplus \text{Box}\left(-CNF\mathbf{V}\right) \oplus \text{Box}(F\mathbf{V}). \tag{41}$$

When $k \geq 2$, the interval estimation of ε_k can be gained as

$$\begin{aligned} \left[\overline{\varepsilon}_k, \underline{\varepsilon}_k\right] = {} & \text{Box}\left(C\left(\bar{A}\right)^k \Pi^a_0\right) \oplus \text{Box}\left(-CNF\mathbf{V}\right) \oplus \text{Box}\left(F\mathbf{V}\right) \\ & \oplus \bigoplus_{i=0}^{k-2} \text{Box}\left(C\left(\bar{A}\right)^i \left[-\bar{A}NF - L_aF\right]\mathbf{V}\right) \\ & \oplus \text{Box}\left(-C\left(\bar{A}\right)^{k-1} L_aF\mathbf{V}\right). \end{aligned} \tag{42}$$

Proof: From (37), when $k \geq 2$, the residual ε_k can be shown as

$$\begin{aligned} \varepsilon_k = {} & C\bar{A}^k e_0 - \sum_{i=0}^{k-1} C\overline{A}^i L_aFv_{k-i-1} + Fv_k \\ & - \sum_{i=0}^{k-1} C\overline{A}^i NFv_{k-i}. \end{aligned} \tag{43}$$

When $k = 1$, the residual ε_1 can be expressed as

$$\varepsilon_1 = C\overline{A}e_0 - CL_aFv_0 + Fv_1 - CNFv_1. \tag{44}$$

When $k = 0$, the residual ε_0 can be gained as

$$\varepsilon_0 = Ce_0 + Fv_0. \tag{45}$$

The reachability set of the residual ε_k is defined as $\varkappa^\varepsilon_k$. Based on $x_0 \in \langle p_0, H^a_0 \rangle$ and $\hat{x}_0 = p_0$, we can get $e_0 = x_0 - \hat{x}_0 \in \langle 0, H^a_0 \rangle$. Then, when $k = 0$, $\varkappa^\varepsilon_0$ can be obtained as

$$\varkappa^\varepsilon_0 = C\langle 0, H^a_0 \rangle \oplus F\mathbf{V}. \tag{46}$$

When $k = 1$, $\varkappa^\varepsilon_1$ can be described as

$$\varkappa^\varepsilon_1 = C\overline{A}\langle 0, H_0 \rangle \oplus (-CL_aF)\mathbf{V} \oplus (-CNF)\mathbf{V} \oplus F\mathbf{V}. \tag{47}$$

When $k \geq 2$, $\varkappa^\varepsilon_k$ can be achieved as

$$\begin{aligned} \varkappa^\varepsilon_k = {} & C\left(\bar{A}\right)^k \langle 0, H_0 \rangle \oplus (-CNF)\mathbf{V} \oplus F\mathbf{V} \\ & \oplus \bigoplus_{i=0}^{k-2} C\left(\bar{A}\right)^i \left[-\bar{A}NF - L_aF\right]\mathbf{V} \\ & \oplus \left(-C\left(\bar{A}\right)^{k-1} L_aF\right)\mathbf{V}. \end{aligned} \tag{48}$$

Subsequently, the interval estimation of the system residual ε_k is obtained, such that $[\overline{\varepsilon}_k, \underline{\varepsilon}_k] = \text{Box}(\varkappa^\varepsilon_k)$. When the system is attack-free, $\overline{\varepsilon}_k < \varepsilon_k < \underline{\varepsilon}_k$ is satisfied. □

Based on the above discussion, we present Algorithm 1 to calculate the interval estimation of the residual ε_k, such that $[\overline{\varepsilon}_k, \underline{\varepsilon}_k]$. If the residual exceeds the boundary, an alarm from the attack detection mechanism will be triggered. The proposed attack detection method based on the system residual is presented as

$$\begin{cases} \varepsilon_k \in \left[\underline{\varepsilon}_k, \overline{\varepsilon}_k\right], & \text{attack-free} \\ \varepsilon_k \notin \left[\underline{\varepsilon}_k, \overline{\varepsilon}_k\right], & \text{attacked.} \end{cases} \tag{49}$$

Remark 3: The proposed method achieves high computational efficiency through an iterative algorithm with linear complexity, compact zonotope representations that enable fast set operations, and a decoupling method that reduces conservatism. Compared to traditional methods in [4], [35], and [36], it eliminates cooperative constraints while maintaining accuracy, making it suitable for real-time applications.

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Algorithm 1 Attack Detection Threshold Generation Based on the System Residual

Require: u_k, y_k
Ensure: $\underline{\varepsilon}_k$, $\bar{\varepsilon}_k$
Initialization:
$\hat{x}_0 = p_0$, $T_x^0 = \langle 0, H_0 \rangle$, $T_v^0 = \emptyset$
for $k \geq 0$ **do**
  $[\underline{r}_k, \bar{r}_k] = \text{Box}(CT_x^k) \oplus T_a \oplus T_b \oplus T_c \oplus T_v^k$
  $T_x^{k+1} = \bar{A} T_x^k$
  $T_a = \text{Box}(-CNF)\mathbf{V}$
  $T_b = \text{Box}(F)\mathbf{V}$
  $T_c = \text{Box}(-C(\bar{A})^{k-1} L_a F)\mathbf{V}$
  $T_v^{k+1} = T_v^k \oplus \text{Box}(C(\bar{A})^i(-\bar{A})NF\mathbf{V})$
    $\oplus \text{Box}(C(\bar{A})^i(-L_aF)\mathbf{V})$
end for

The reachability-based attack detection further reduces the computational burden by replacing complex geometric checks with efficient interval arithmetic.

C. Attack Reconstruction Based on Zonotopes

In this section, FDI attacks can be estimated using the system state x_k observed from the UIO. According to the definition of the error system (22), the system state x_k can be obtained as $x_k = e_k + \hat{x}_k$, where $e_k \in \langle p_k^a, H_k^a \rangle$.

Theorem 4: Consider the system (10) under FDI attacks. Then, FDI attacks can be reconstructed as

$$\hat{a}_k^u = D_u^\dagger \left(p_k^a + \hat{x}_k - A\left(p_{k-1}^a + \hat{x}_{k-1} \right) - Bu_{k-1} \right) \tag{50}$$
$$\hat{a}_k^y = D_y^\dagger \left(y_k - C\left(p_k^a + \hat{x}_k \right) \right) \tag{51}$$

where

$$\hat{a}_k^u \in \left\langle p_k^u, H_k^u \right\rangle, \quad \hat{a}_k^y \in \left\langle p_k^y, H_k^y \right\rangle$$

with

$$\begin{aligned} p_k^u &= D_u^\dagger \left(p_k^a + \hat{x}_k - A\left(p_{k-1}^a + \hat{x}_{k-1} \right) - Bu_{k-1} \right) \\ H_k^u &= \begin{bmatrix} D_u^\dagger H_k^a & D_u^\dagger A H_{k-1}^a & D_u^\dagger E H_w \end{bmatrix} \\ p_k^y &= D_y^\dagger \left(y_k - C\left(p_k^a + \hat{x}_k \right) \right) \\ H_k^y &= \begin{bmatrix} D_y^\dagger C H_k^a & D_y^\dagger F H_v \end{bmatrix}. \end{aligned}$$

We assume that $\hat{\mathcal{A}}_k^u$ and $\hat{\mathcal{A}}_k^y$ represent the sets of isolated actuator and sensor attacks, respectively. The isolation of FDI attacks can be accomplished in accordance with

$$\hat{\mathcal{A}}_k^u = \text{supp}\left(\tau_{k,i}^u \right), \quad \hat{\mathcal{A}}_k^y = \text{supp}\left(\tau_{k,i}^y \right) \tag{52}$$

where $\tau_{k,i}^u = 0$ if $\hat{a}_{k,i}^u \in \langle 0, H_k^u \rangle$; otherwise, $\tau_{k,i}^u = 1$. $\tau_{k,i}^y = 0$ if $\hat{a}_{k,i}^y \in \langle 0, H_k^y \rangle$; otherwise, $\tau_{k,i}^y = 1$.

Proof: Based on the definition of the error system, we can derive

$$e_{k+1} + \hat{x}_{k+1} = A\left(e_k + \hat{x}_k \right) + Bu_k + D_u a_k^u + Ew_k.$$

Then, one can calculate

$$a_k^u = D_u^\dagger \left(e_{k+1} - Ae_k + \hat{x}_{k+1} - A\hat{x}_k - Ew_k - u_k \right). \tag{53}$$

Algorithm 2 Attack Reconstruction and Isolation

Require: A, B, C, p_k^a, y_k, and $\hat{x}_k$
while $k \leq k_N$ **do**
  $\hat{a}_k^u = D_u^\dagger (p_k^a + \hat{x}_k - A(p_{k-1}^a + \hat{x}_{k-1}) - Bu_{k-1})$
  $\hat{a}_k^y = D_y^\dagger (y_k - C(p_k^a + \hat{x}_k))$
  $\hat{\mathcal{A}}_k^u = \text{supp}(\tau_{k,i}^u)$
  $\hat{\mathcal{A}}_k^y = \text{supp}(\tau_{k,i}^y)$
end while
Ensure: $\hat{a}_k^u$, $\hat{a}_k^y$, $\hat{\mathcal{A}}_k^u$, and $\hat{\mathcal{A}}_k^y$

Similarly, from the system (10) in the presence of FDI attacks, we can obtain

$$y_k = C\left(e_k + \hat{x}_k \right) + Fv_k + D_y a_k^y. \tag{54}$$

Therefore, one can get

$$a_k^y = D_y^\dagger \left(y_k - C\left(e_k + \hat{x}_k \right) - Fv_k \right). \tag{55}$$

In practical systems, it is pointed out that the error system, disturbances, and measurement noise cannot be precisely measured but can be limited in a zonotope, respectively. The reconstruction of attacks can be shown as

$$\hat{a}_k^u = D_u^\dagger \left(p_k^a + \hat{x}_k - A\left(p_{k-1}^a + \hat{x}_{k-1} \right) - Bu_{k-1} \right) \tag{56}$$
$$\hat{a}_k^y = D_y^\dagger \left(y_k - C\left(p_k^a + \hat{x}_k \right) \right). \tag{57}$$

Based on (8) and (37), one can get

$$\hat{a}_k^u \in \left\langle p_k^u, H_k^u \right\rangle, \quad \hat{a}_k^y \in \left\langle p_k^y, H_k^y \right\rangle$$

where the specific expressions of p_k^u, H_k^u, p_k^y, and H_k^y are given in upon discussion. The reconstruction of actuator attack $\hat{a}_k^u$ is selected based on (50). Then, if the ith actuator is attack-free, we can get that $\hat{a}_{k,i}^u \in \langle 0, H_k^u \rangle$. On the other hand, if the ith actuator is attacked, the condition needs to satisfy $\hat{a}_{k,i}^u \notin \langle 0, H_k^u \rangle$. Now, the isolation of the actuator attacks is represented as

$$\begin{cases} \tau_{k,i}^u = 0, & \text{attack-free} \\ \tau_{k,i}^u = 1, & \text{attacked.} \end{cases}$$

Similarly, the isolation of the sensor attacks can be accomplished too. The proof is completed. □

Theorem 4 investigates the impact of disturbances and measurement noise on the reconstruction of the attacks. All attacks can be thoroughly reconstructed if the disturbances and measurement noises are not considered.

The reconstruction and isolation of FDI attacks utilizing the proposed UIO can be completed in accordance with the following Algorithm 2.

Remark 4: Unlike the existing methods proposed in [45], [46], and [47], our work primarily focuses on attack detection and reconstruction within the framework of discrete-time linear systems. In future work, we will explore extending our approach to linear parameter-varying (LPV) systems.

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IV. Simulation Study

In this section, we consider the following the discrete-time linear system (8) in the presence of FDI attacks

$$A = \begin{pmatrix} -0.54 & 0.45 & 0.36 & 0 & 0 \\ 0.63 & 0.45 & 0.18 & 0.36 & 0 \\ 0.09 & 0.45 & 0.27 & 0.09 & 0.18 \\ 0 & 0 & 0.25 & 0.25*\sqrt{2} & -0.1*\sqrt{2} \\ 0 & 0 & 0 & 0.25*\sqrt{2} & 0.25*\sqrt{2} \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad E = \begin{pmatrix} 0.1 \\ 0 \\ 0 \\ 0 \\ 0.1 \end{pmatrix}$$

$$F = \begin{pmatrix} 0.1 & 0 \\ 0 & 0.1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

In this case, the disturbances and measurement noise are set to satisfy the following condition: $|w_k| \leq 0.8$ and $|v_k| \leq [0.5\ 0.5]^{\mathrm{T}}$, respectively. The initial system state is set to $x_0 = [4\ \ -4\ \ 4\ \ -4\ \ 4]^T$. The input u_k is chosen as

$$u_k = \begin{pmatrix} 0.1\sin(0.1k) \\ 0.1\cos(0.2k) \\ 0.1\sin(0.1k) \\ 0.1\cos(0.2k) \\ 0.1\sin(0.1k) \end{pmatrix}, \quad k \in \mathbb{Z}^+.$$

Then, the values of all elements of the matrix N and L_a can be calculated by (19) and (28) as

$$N = \begin{pmatrix} 1 & 0.0052 \\ 0 & 0.1227 \\ 0 & 0.0146 \\ 0 & 0.3119 \\ 1 & -0.6075 \end{pmatrix}, \quad L_a = \begin{pmatrix} 0.0017 & -0.0022 \\ 0.4536 & 0.0845 \\ 0.2819 & -0.0974 \\ 0.2382 & 0.2756 \\ 0.8227 & -0.3547 \end{pmatrix}.$$

It is supposed that the sensors and actuators can be attacked, i.e., $D_u \neq 0$ and $D_y \neq 0$. Then, we choose $D_u = \mathrm{diag}\{1, 1, 1, 1, 1\}$ and $D_y = \mathrm{diag}\{1, 1\}$, which satisfies assumption 2. In addition, FDI attacks were selected as

$$a_k^u = \begin{pmatrix} 0 \\ 1 \\ 0.02k_a \\ 0 \\ 0 \end{pmatrix}, \quad a_k^y = \begin{pmatrix} 0.05k_a \\ 0 \end{pmatrix}, \quad k_a \in [15, 40].$$

The simulation results are displayed in Figs. 1–8. Figs. 1 and 2 are obtained by solving Theorem 3 and demonstrate the attack detection approach based on the bounds of the system residual. Figs. 1 and 2 compare the bounds of the system residual, calculated using our proposed method, with those obtained without employing the decoupling method. The results show that the system residual remains within the calculated bounds when the system is attack-free and breaches these bounds when an attack occurs, demonstrating the effectiveness of our attack detection approach.

The results of attacks reconstruction based on Theorem 4 are shown in Figs. 3–6. Figs. 3–5 represent the reconstruction of the actuator attacks a_k^u. Fig. 6 denotes the reconstruction of the sensor attacks a_k^y. Figs. 3–6 clearly show that the reconstructed attack signals closely follow the actual attack signals,

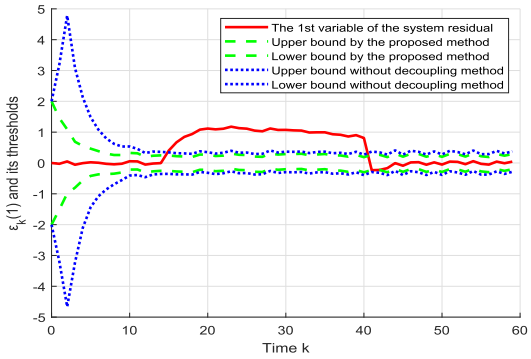


Fig. 1. First system residual and its thresholds.

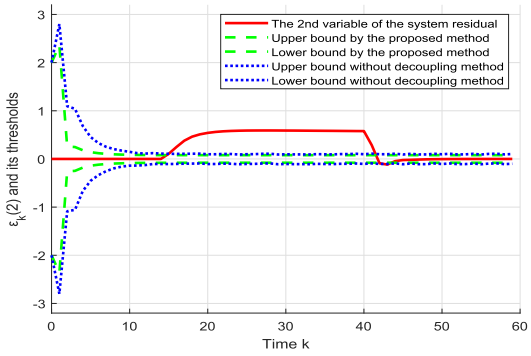


Fig. 2. Second system residual and its thresholds.

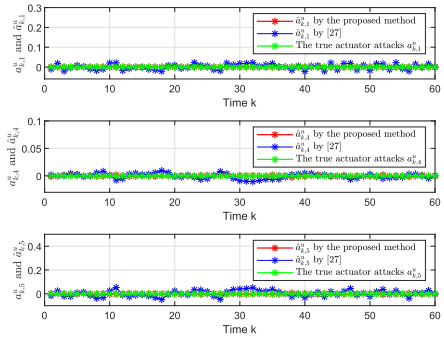


Fig. 3. Reconstruction of actuators attacks $a_{k,1}^u$, $a_{k,4}^u$, and $a_{k,5}^u$.

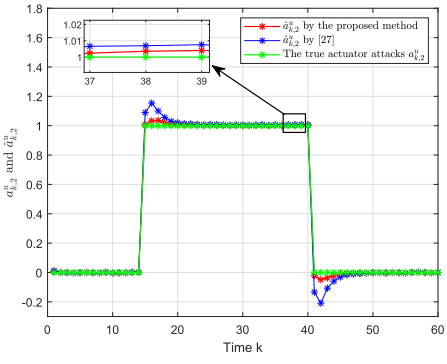


Fig. 4. Reconstruction of actuator attack $a_{k,2}^u$.

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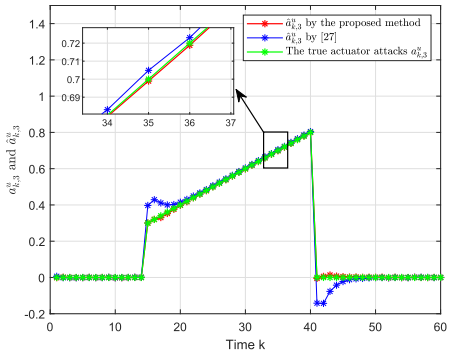


Fig. 5. Reconstruction of actuator attack $a^u_{k,3}$.

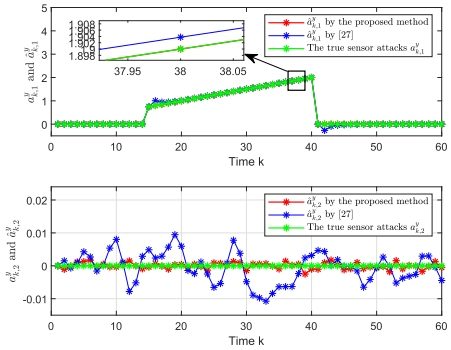


Fig. 6. Reconstruction of sensor attacks $a^y_{k,1}$ and $a^y_{k,2}$.

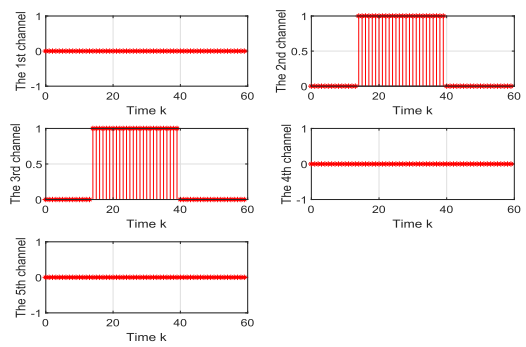


Fig. 7. Attack isolation of a^u_k.

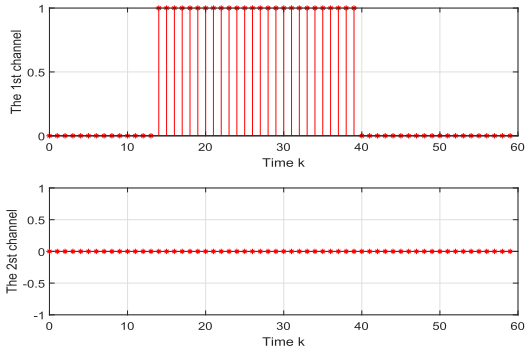


Fig. 8. Attack isolation of a^y_k.

indicating the high accuracy of our proposed reconstruction method.

The isolation of FDI attacks a^u_k and a^y_k is shown in Figs. 7 and 8, respectively. Figs. 7 and 8 demonstrate that our method can effectively isolate the attacks on the actuator and sensor components, with a clear indication of the time intervals when the system is attacked.

V. Conclusion

In this work, we have proposed an improved attacks detection and reconstruction approach for discrete-time linear CPSs with unknown-but-bounded disturbances and measurement noise, which has been developed by integrating the UIO with reachability analysis. Through the application of the decoupling method, the effect of the disturbances from the error system could be excluded. Then, compared with the existing methods, the proposed attack detection method based on the reachability analysis of the system residual requires less computation, which is more accuracy. Moreover, the effect of disturbances and measurement noise on attack reconstruction and isolation were investigated. The accuracy of the reconstruction of FDI attacks was enhanced by decoupling method and H_∞ technology. Finally, the simulation results verified the effectiveness and feasibility of the proposed approach. In the future, we will further explore the extension of the proposed method to LPV systems and combine scheduling strategies and reconstruction mechanisms to realize the design of resilient systems [48], [49], [50], [51].

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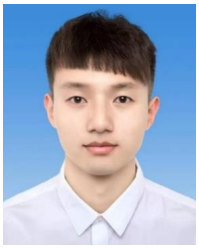
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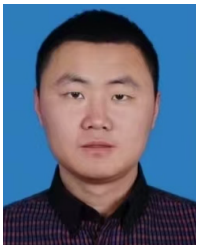
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